

GYMGEEK

AI-INTENSIVE FITNESS GUIDANCE SYSTEM



System Documentation

Engineering of AI-Intensive Systems

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Project documentation covering requirements, use cases, domain model, architecture, implementation, and design decisions.

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Document conventions

- FR denotes a functional requirement; NFR denotes a non-functional requirement.
- UC denotes a main use case; SUC denotes a supporting use case.
- AI components are identified explicitly in the domain and architecture models.

1. System Goal and Scope

GymGeek is a mobile fitness guidance application intended for gym users who need immediate support when selecting and using equipment. The system combines camera-based equipment recognition, instructional content, workout logging, progress tracking, research-grounded fitness summaries, and goal-based recommendations in one Flutter application.

The primary objective is to help users use gym equipment safely and effectively. A user can point the phone camera at a machine, review the predicted equipment class and confidence, open the matching equipment record, and continue to instructions or workout logging. Manual search remains available when camera recognition is unsuitable.

1.1 Target Users

- Beginners who are unfamiliar with machine names and operating procedures.
- Intermediate and advanced gym users who want concise technique guidance.
- Amateur and professional athletes who track sessions and progress.
- Users pursuing general fitness, strength, endurance, or weight-management goals.

1.2 Primary Goals

- Identify gym machines and equipment from a smartphone camera image.
- Present equipment descriptions, safety tips, and instructional videos.
- Summarise selected fitness research in accessible language and display the supporting sources.
- Record workout sessions and sets, then present progress indicators and personal records.
- Prioritise equipment and exercises according to the user's selected fitness goal.

1.3 System Boundary

The prototype covers the Android-oriented Flutter client, bundled equipment and exercise catalogues, an on-device TensorFlow Lite classifier, local SQLite persistence, and a network-based generative summarisation service. Raw camera images are processed during inference and are not stored by the application. External video platforms provide instructional content through URLs recorded in the equipment catalogue.

2. Requirements

The requirements define the expected user-visible behaviour, runtime properties, data controls, and AI-specific quality constraints. They use stable identifiers so that use cases and architecture components can be traced back to the original specification.

2.1 Stakeholders

Stakeholder group	Interest or responsibility
Primary users	Beginner, intermediate, and advanced gym users; athletes; users pursuing general fitness goals.
Gym staff and trainers	May use the application as a supplementary explanation tool and are affected by the quality of the safety guidance.
Gym owners	Interested in user flow, equipment usage, and safe operation of machines.
Development team	Responsible for implementation, model integration, testing, and maintenance.
Course instructors	Evaluate whether the prototype and documentation satisfy course objectives.
Regulatory and consumer-protection bodies	Concerned with privacy, transparency, misleading claims, and safe presentation of fitness information.

2.2 Functional Requirements

ID	Requirement
FR-01	The system shall allow users to authenticate with a user account.
FR-02	After authentication, the system shall display the home interface and access to recent activity.
FR-03	While the camera is open, the system shall process captured images to identify gym equipment.
FR-04	When camera permission is available, the system shall activate the camera and display a live preview.
FR-05	After successful identification, the system shall display the equipment name, description, and confidence information.
FR-06	When the user requests instructions, the system shall retrieve and open an instructional video for the selected equipment.
FR-07	When the user submits a fitness question, the system shall generate a simplified, research-grounded summary.
FR-08	When a workout session or set is completed, the system shall record the activity in the local training history.

ID	Requirement
FR-09	The dashboard shall display progress metrics, including frequency, duration, equipment usage, recent sets, streaks, and records.
FR-10	If no reliable equipment result is available for 15 seconds, the system shall suggest changing the camera position or lighting.
FR-11	The system shall allow equipment to be searched by name, description, muscle group, or model label.
FR-12	The system shall prioritise equipment and exercises that correspond to the user's selected fitness goal.

2.3 Non-Functional Requirements

ID	Requirement
NFR-01	Camera functions shall use the device camera interface supported by the mobile operating system.
NFR-02	Instructional video content shall be opened over HTTPS through an external streaming service.
NFR-03	Workout history, goals, preferences, and audit data shall be stored locally.
NFR-04	AI components shall communicate through defined inference interfaces and structured request/response data.
NFR-05	Initial model loading shall complete within five seconds on the target device.
NFR-06	The recognition function shall be responsive enough for interactive camera use, with a target processing rate of at least 15 frames per second for continuous implementations.
NFR-07	Equipment recognition shall remain available without an internet connection through an on-device model.
NFR-08	AI predictions, summaries, recommendations, confidence values, and user corrections shall be auditable.
NFR-09	The client shall be implemented with a cross-platform framework that supports Android and iOS.
NFR-10	On-device inference shall be preferred where feasible to reduce latency and limit transmission of personal data.
NFR-11	The handling and deletion of user data shall follow GDPR-oriented data minimisation and user-control principles.
NFR-12	Users shall be able to disable AI-powered personalised recommendations.

2.4 AI-Specific Quality Constraints

Area	Constraint
Equipment recognition	Top-3 accuracy target of at least 75%; results displayed with confidence and correction/fallback controls.
Research summarisation	Retrieved sources are supplied as context; generated text is constrained to the retrieved material and is accompanied by citations and a disclaimer.

Area	Constraint
Recommendations	Goal-based recommendations remain advisory, can be disabled, and preserve basic application functionality when disabled.
Safety and transparency	Unsafe terms are filtered, medical claims are avoided, AI output is labelled, and audit events are stored locally.
Privacy	Recognition runs on-device and raw camera images are not retained. User data can be deleted from the profile settings.

3. Use Case Model

The use case model separates user-facing actions from supporting services. Eight main use cases describe the interaction flow, while seven supporting use cases capture authentication, inference, content retrieval, persistence, and recommendation operations.

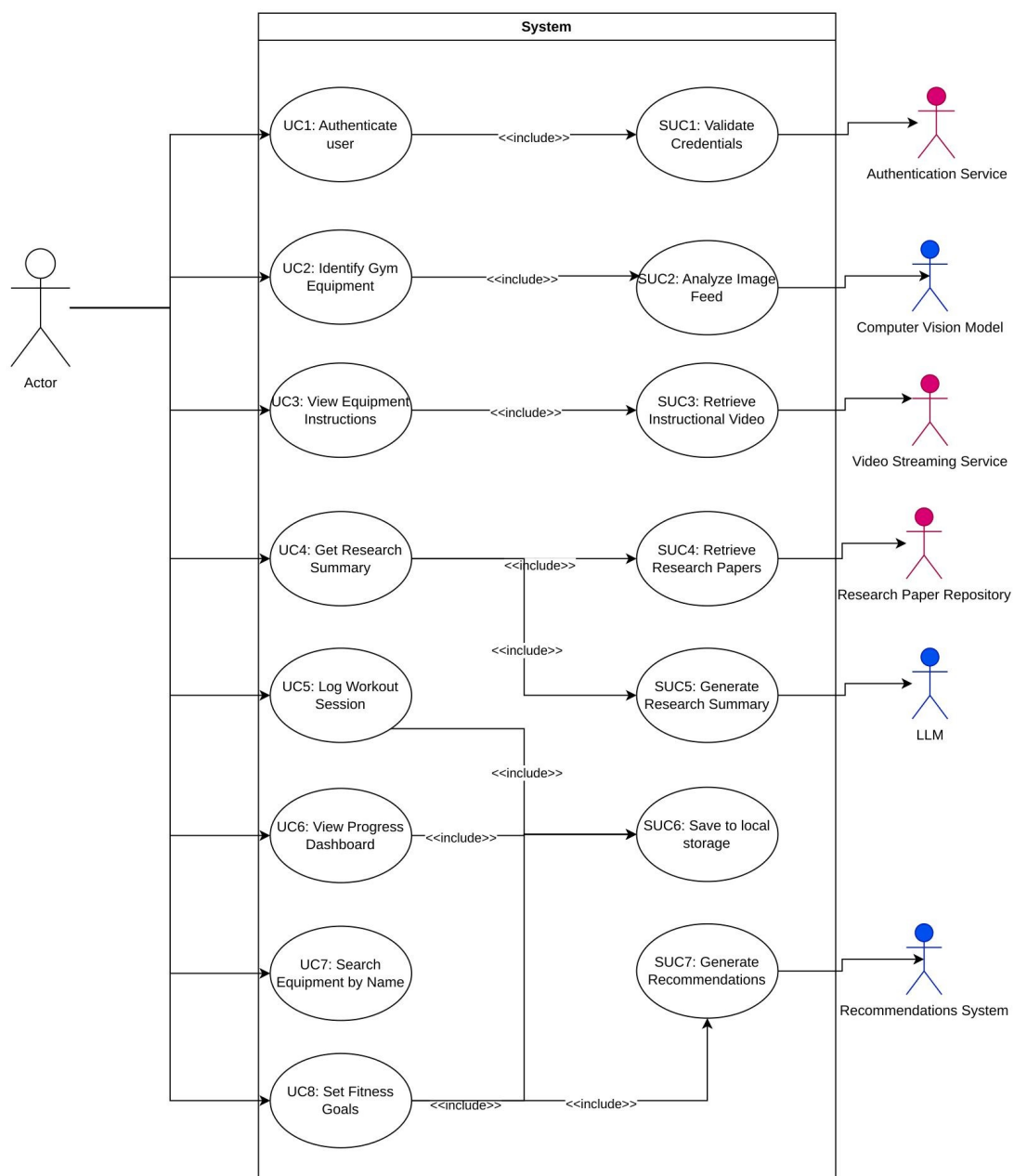


Figure 1. Main and supporting use case diagram.

3.1 UC1: Authenticate User

Description	User logs into the system to access personalized features and training history
Actors	User, Authentication Service
Stakeholders	User, Project development team, Course instructors
Preconditions	User has a registered account; Device has internet connection
Success condition	User is authenticated and redirected to home screen
Failure condition	Authentication fails; User remains on login screen

Main success scenario

Step	Description	Linked UC
1	User opens the application	
2	User enters username and password	
3	System sends credentials to Authentication Service	SUC1
4	Authentication Service validates credentials	
5	System displays home screen with recent equipment and workout history	

Alternative scenarios

Step	Description	Linked UC
4.A1	Authentication Service returns "invalid credentials"	
4.A2	System displays error message: "Incorrect username or password"	
4.A3	System prompts user to try again	
4.A4	Go back to step 2	

Exception scenarios

Step	Description	Linked UC
3.A1	Authentication Service does not respond within 10 seconds	
3.A2	System displays error message: "Connection timeout. Please try again"	
3.A3	System remains on login screen	

3.2 UC2: Identify Gym Equipment

Description	User captures gym equipment using camera and receives identification results
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Actors	User, Camera, Computer Vision Model
Stakeholders	User, Gym staff, Gym owners, Project development team
Preconditions	User has granted camera permissions; Camera is functional
Success condition	Equipment is identified correctly; Name and description are displayed
Failure condition	Equipment cannot be identified after multiple attempts

Main success scenario

Step	Description	Linked UC
1	User opens camera view from home screen	
2	System activates camera feed and displays live preview	
3	User points camera at gym equipment	
4	System processes image feed to identify equipment	SUC2
5	Computer Vision Model returns identification results	
6	System displays equipment name and brief description	
7	User confirms the identification is correct	

Alternative scenarios

Step	Description	Linked UC
5.A1	Computer Vision Model returns identification with confidence below 75%	
5.A2	System displays "Low confidence. Please adjust camera position"	
5.A3	User adjusts camera angle	
5.A4	Go back to step 4	

Exception scenarios

Step	Description	Linked UC
4.A1	Computer Vision Model takes more than 15 seconds to return a result	
4.A2	System displays error message: "Processing timeout. Please try again"	
4.A3	System keeps camera open for new attempt	
5.A1	No equipment detected for 15 seconds while camera is open	
5.A2	System displays message: "No equipment detected. Ensure proper lighting and camera position"	

3.3 UC3: View Equipment Instructions

Description	User retrieves and watches instructional video for identified equipment
Actors	User, Video Streaming Service
Stakeholders	User, Gym staff, Project development team
Preconditions	Equipment has been successfully identified (UC-02 completed); Internet connection available
Success condition	Instructional video plays correctly showing equipment usage
Failure condition	Video cannot be retrieved or played

Main success scenario

Step	Description	Linked UC
1	User selects "View Instructions" after equipment identification	
2	System retrieves instructional video for detected equipment	SUC3
3	Video Streaming Service returns video content	
4	System plays video showing correct equipment usage	
5	User watches video	

Alternative scenarios

Step	Description	Linked UC
2.A1	Multiple videos exist for the same equipment type	
2.A2	System displays a list of available videos	
2.A3	User selects preferred video from list	
2.A4	Go back to step 4	

Exception scenarios

Step	Description	Linked UC
2.A1	No instructional video exists for detected equipment	
2.A2	System displays message: "No video available for this equipment"	
2.A3	System suggests searching by equipment name instead	
3.A1	Video fails to load within 10 seconds	
3.A2	System displays message: "Video loading failed. Check your connection"	
3.A3	System offers to retry loading	

3.4 UC4: Get Research Summary

Description	User requests simplified summary of scientific fitness research on a topic
Actors	User, Large Language Model (LLM), Research Paper Repository
Stakeholders	User, Researchers, Project development team, Consumer protection agencies
Preconditions	Internet connection available; User has typed a search query
Success condition	Simplified research summary is displayed to user
Failure condition	No summary can be generated or retrieved

Main success scenario

Step	Description	Linked UC
1	User navigates to research summary feature	
2	User types a fitness-related question or topic	
3	System sends query to Research Paper Repository	SUC4
4	Research Paper Repository returns relevant papers	
5	System sends papers to LLM for summarization	SUC5
6	LLM generates simplified summary	
7	System displays summary to user with source references	

Alternative scenarios

Step	Description	Linked UC
3.A1	Research Paper Repository returns no relevant papers	
3.A2	System displays message: "No research found for this topic. Try different keywords"	
3.A3	User modifies search query	
3.A4	Go back to step 2	

Exception scenarios

Step	Description	Linked UC
5.A1	LLM takes more than 15 seconds to generate complete response	
5.A2	System displays partial summary with note: "Generation in progress..."	
5.A3	If no response after 20 seconds, system displays timeout message	

Step	Description	Linked UC
6.A1	LLM generates summary containing potentially unsafe advice	
6.A2	Safety guardrail flags the content	
6.A3	System displays disclaimer: "This advice requires professional consultation"	

3.5 UC5: Log Workout Session

Description	User saves completed workout session data to training history
Actors	User, Local Storage
Stakeholders	User, Project development team
Preconditions	User is authenticated; User has completed a workout
Success condition	Workout session data is saved to user's history
Failure condition	Session data cannot be saved

Main success scenario

Step	Description	Linked UC
1	User completes a workout session	
2	User selects "Save Session" or session auto-saves	
3	System collects session data (equipment used, duration, timestamp)	
4	System saves session data to local storage	SUC6
5	System confirms "Workout saved successfully"	

Alternative scenarios

Step	Description	Linked UC
4.A1	Local storage is full	
4.A2	System displays message: "Storage full. Delete old sessions to save new ones"	
4.A3	User deletes old sessions	
4.A4	Go back to step 4	

Exception scenarios

Step	Description	Linked UC
4.A1	Write operation to local storage fails	
4.A2	System displays error message: "Failed to save session. Please try again"	

Step	Description	Linked UC
4.A3	System retains session data in memory for retry	
4.A4	User attempts to save again	

3.6 UC6: View Progress Dashboard

Description	User views dashboard displaying workout statistics and progress metrics
Actors	User
Stakeholders	User, Project development team
Preconditions	User is authenticated; User has saved workout history (UC-05)
Success condition	Dashboard displays accurate progress metrics
Failure condition	Dashboard cannot load data

Main success scenario

Step	Description	Linked UC
1	User selects "Dashboard" from navigation menu	
2	System retrieves user's training history from local storage	SUC6
3	System calculates progress metrics (frequency, duration, equipment usage)	
4	System displays dashboard with charts and statistics	

Alternative scenarios

Step	Description	Linked UC
2.A1	User has no saved workout history	
2.A2	System displays message: "No workout data yet. Complete a workout to see statistics"	
2.A3	System shows placeholder dashboard with encouragement message	

Exception scenarios

Step	Description	Linked UC
2.A1	Local storage is corrupted or inaccessible	
2.A2	System displays error message: "Cannot load workout data"	
2.A3	System prompts user to restart application	

3.7 UC7: Search Equipment by Name

Description	User searches for equipment using text input instead of camera
Actors	User
Stakeholders	User, Project development team
Preconditions	User is authenticated (or using guest mode); Internet connection available
Success condition	Matching equipment is displayed with instructional content
Failure condition	No matching equipment found

Main success scenario

Step	Description	Linked UC
1	User selects search feature	
2	User types equipment name into search bar	
3	System searches equipment database for matches	SUC3
4	System returns matching equipment with instructional content	
5	User selects equipment from results	

Alternative scenarios

Step	Description	Linked UC
3.A1	Search returns partial matches (e.g., "leg" matches "leg press", "leg extension")	
3.A2	System displays all partial matches ranked by relevance	
3.A3	User selects from the list	
3.A4	Go back to step 5	

Exception scenarios

Step	Description	Linked UC
3.A1	Search query contains typos or misspellings	
3.A2	System suggests: "Did you mean: [corrected spelling]"	
3.A3	User accepts suggestion or modifies search	
4.A1	No equipment matches the search query	
4.A2	System displays message: "No equipment found. Try different keywords or use camera"	

3.8 UC8: Set Fitness Goals

Description	User indicates fitness goals to receive personalized recommendations
Actors	User, Recommendation System
Stakeholders	User, Project development team
Preconditions	User is authenticated
Success condition	User's goals are saved and used for personalization
Failure condition	Goals cannot be saved

Main success scenario

Step	Description	Linked UC
1	User navigates to "Goals" section in profile	
2	User selects fitness goals (e.g., strength, endurance, weight loss)	
3	System saves selected goals to user profile	SUC6
4	System sends goals to Recommendation System	SUC7
5	Recommendation System updates personalization model	
6	System confirms "Goals saved successfully"	

Alternative scenarios

Step	Description	Linked UC
3.A1	User modifies existing goals	
3.A2	System updates goals in profile (overwrites previous)	
3.A3	Go back to step 6	

Exception scenarios

Step	Description	Linked UC
3.A1	Save operation to local storage fails	
3.A2	System displays error message: "Failed to save goals. Please try again"	
3.A3	User retries saving	

3.9 Supporting Use Cases

ID	Supporting use case	Description	Primary actor/component
SUC1	Validate Credentials	Authentication service validates user credentials	Authentication Service

ID	Supporting use case	Description	Primary actor/component
SUC2	Analyze Image Feed	Computer Vision Model analyzes camera feed to identify gym equipment	Computer Vision Model
SUC3	Retrieve Instructional Video	System retrieves instructional video for identified or searched equipment	Video Streaming Service
SUC4	Retrieve Research Papers	Research Paper Repository retrieves relevant papers based on user query	Research Paper Repository
SUC5	Generate Research Summary	Large Language Model generates simplified summary of research papers	Large Language Model (LLM)
SUC6	Save to Local Storage	System saves user data to local device storage	Local Storage
SUC7	Generate Recommendations	Recommendation System generates personalized exercise recommendations	Recommendation System

3.10 Implemented Use Cases

ID	Use case	Status	Primary source files	Implementation note
UC1	Authenticate User	Implemented	login_screen.dart, main.dart	Validates the prototype account, handles errors and timeout, and persists login state.
UC2	Identify Gym Equipment	Implemented	camera_screen.dart, detection_service.dart	Uses the live camera, periodic capture, on-device TFLite inference, confidence threshold, top alternatives, feedback, and manual-search fallback.
UC3	View Equipment Instructions	Implemented	video_player_screen.dart, equipment.json	Displays equipment details and opens one or more instructional videos through the device browser or video application.
UC4	Get Research Summary	Implemented	research_screen.dart, research_service.dart	Retrieves relevant local research records, generates a concise summary, displays sources, and provides a research-excerpt fallback.
UC5	Log Workout Session	Implemented	video_player_screen.dart, exercise_detail_screen.dart, database_service.dart	Stores sessions, sets, repetitions, weight, notes, and personal-record information in SQLite.
UC6	View Progress Dashboard	Implemented	dashboard_screen.dart, database_service.dart	Aggregates recent activity, total duration, weekly workouts, streaks, equipment usage, and recent sets for chart-based presentation.
UC7	Search Equipment by Name	Implemented	equipment_search_screen.dart, exercise_library_screen.dart	Searches the local catalogues by equipment name, description, muscle group, and aliases.
UC8	Set Fitness Goals	Implemented	profile_screen.dart, goal.dart, equipment_service.dart	Stores the current goal and derives content-based equipment and exercise suggestions from relevant muscle groups.

3.11 Traceability Matrix

An X indicates that the requirement is realised or supported by the corresponding main or supporting use case.

Requirement	UC1	UC2	UC3	UC4	UC5	UC6	UC7	UC8	SUC1	SUC2	SUC3	SUC4	SUC5	SUC6	SUC7
FR-01	X								X						
FR-02	X					X									
FR-03		X								X					
FR-04		X													
FR-05		X								X					
FR-06			X				X				X				
FR-07				X								X	X		
FR-08					X									X	
FR-09						X									
FR-10		X													
FR-11							X								
FR-12								X							X
NFR-01	X	X													
NFR-02			X								X				
NFR-03					X	X		X						X	
NFR-04		X		X						X		X	X		X
NFR-05		X								X					
NFR-06		X								X					
NFR-07		X								X					
NFR-08				X									X		X
NFR-09	X		X	X	X	X	X	X							
NFR-10		X								X					X
NFR-11					X	X		X						X	
NFR-12								X							

4. Domain Model

The domain model defines the persistent and conceptual entities used by the application. AI-specific entities are separated from ordinary application data, but all classes remain connected through user actions, equipment selection, research requests, workout history, and recommendations.

Class	Role in the domain
User	Account identity, profile data, authentication state, goals, research requests, and workout ownership.
FitnessGoal	Selected goal type, target information, and the basis for content-based recommendations.
WorkoutSession	Equipment-based activity with start/end or duration data and optional notes.
Equipment	Machine name, description, muscle group, difficulty, aliases, safety tips, and instructional content.
InstructionalVideo	Title and URL for an equipment-specific demonstration.
ProgressMetrics	Derived statistics such as weekly frequency, total duration, streak, usage counts, and records.
ResearchPaper	Curated title, authors, year, abstract, and keywords used by retrieval.
ResearchSummary	Generated or fallback content, source papers, generation status, and presentation metadata.
CVModel	Model version, confidence, inference time, detection operation, and equipment-class output.
LLM	Generative model configuration, summary generation operation, and safety checking.
RecommendationSystem	Goal- and history-based equipment or exercise prioritisation.

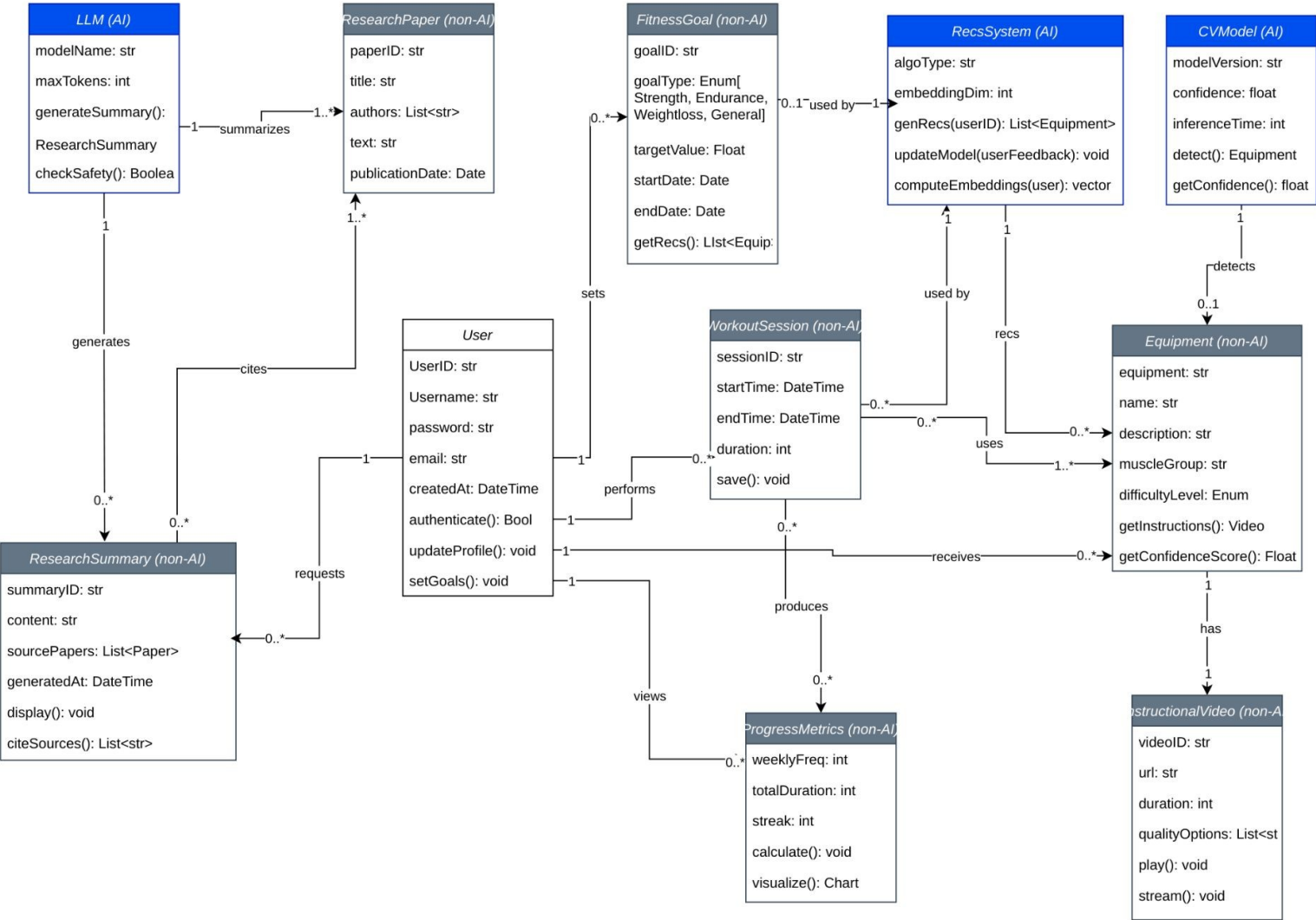


Figure 2. GymGeek domain model. Blue headers denote AI-related classes; grey headers denote non-AI classes.

5. System Architecture

GymGeek uses a layered mobile architecture. Flutter screens provide the user interface, service classes coordinate inference and persistence, bundled JSON files form the local content stores, SQLite keeps user activity, and external services are used only where network access is necessary. The on-device classifier remains independent of the network path used for research summarisation and video access.

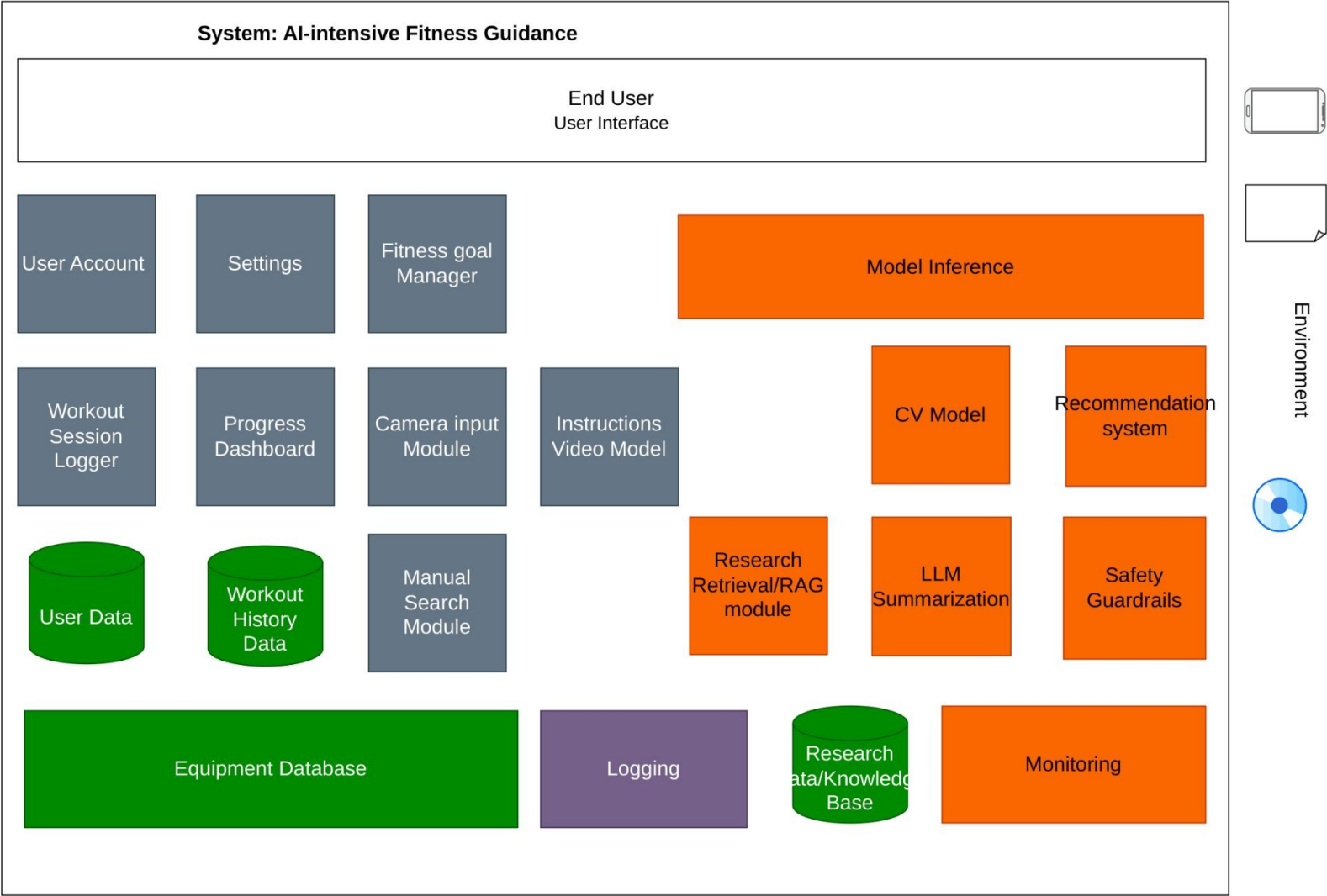


Figure 3. Logical architecture. Grey: non-AI components; orange: AI components; green: data storage; purple: infrastructure.

5.1 Runtime Data Flow

- Equipment recognition: Camera Input -> Detection Service -> TFLite CV Model -> Equipment Catalogue -> Equipment Details and Instructions.
- Research summary: User Query -> Local Retrieval -> Selected Research Records -> Generative Summary -> Safety Check -> Summary with Sources.
- Workout tracking: Equipment or Exercise -> Session/Set Logger -> SQLite -> Dashboard Aggregation and Charts.
- Recommendations: Selected Fitness Goal -> Goal-to-Muscle Mapping -> Filtered Equipment and Exercise Catalogue.
- Auditability: CV detections, user feedback, summary events, goal changes, and workout logging are recorded in the local audit table.

5.2 Component Descriptions

Component	Type	Responsibility	Related requirements
Mobile User Interface	Non-AI	Flutter screens for login, scanning, library access, instructions, research, dashboard, and profile settings.	FR-01, FR-02, FR-05, FR-06, FR-07, FR-09, FR-11, FR-12, NFR-09
User Account Management	Non-AI	Validates the prototype account and stores the authenticated session state.	FR-01, FR-02, NFR-03, NFR-11
Settings / Privacy Control	Non-AI	Stores consent and AI-recommendation preferences and provides user-data deletion.	NFR-11, NFR-12
Fitness Goal Manager	Non-AI	Stores one active goal and supplies the recommendation filter.	FR-12
Camera Input Module	Non-AI	Controls permission, live preview, flash, camera switching, capture timing, and no-detection timeout.	FR-03, FR-04, FR-10, NFR-01
CV Model	AI	ResNet-50 classifier converted to TensorFlow Lite; predicts one of 23 equipment classes.	FR-03, FR-05, FR-10, NFR-05, NFR-07, NFR-10
Model Inference Service	AI	Loads the model and labels, preprocesses 224 x 224 RGB input, runs inference, ranks outputs, applies the threshold, and logs detections.	FR-03, FR-05, FR-10, NFR-04, NFR-05, NFR-07, NFR-08, NFR-10
Equipment Database	Data	Bundled JSON catalogue containing equipment names, aliases, descriptions, muscle groups, tips, and video links.	FR-05, FR-06, FR-11
Instructional Video Module	Non-AI	Presents equipment content and opens one or more external instructional videos.	FR-06, FR-11, NFR-02
Manual Search Module	Non-AI	Searches equipment and exercise records and acts as the recognition fallback.	FR-10, FR-11
Research Retrieval / RAG Module	AI	Scores local research records against the user query and returns the three most relevant documents.	FR-07, NFR-08
Research Knowledge Base	Data	Bundled JSON collection of curated research records with abstracts and retrieval keywords.	FR-07, NFR-08
LLM Summarisation	AI	Generates a short answer using only retrieved context and returns source metadata to the interface.	FR-07, NFR-04, NFR-08
Safety Guardrails	AI	Filters unsafe topics, constrains the prompt, and appends a non-medical disclaimer.	FR-07, FR-12, NFR-08, NFR-11, NFR-12

Component	Type	Responsibility	Related requirements
Recommendation System	AI	Uses the selected goal to filter equipment and exercises by relevant muscle groups.	FR-12, NFR-08, NFR-12
Workout Session Logger	Non-AI	Stores equipment sessions and detailed exercise sets, including weight and repetitions.	FR-08, FR-09, FR-12
Workout History Data	Data	SQLite tables for sessions and sets used by the dashboard and record calculations.	FR-08, FR-09, FR-12, NFR-03
Progress Dashboard	Non-AI	Calculates and visualises duration, frequency, streak, usage, activity history, and personal records.	FR-09
User Data	Data	Local profile state, goals, preferences, consent, and authentication state.	FR-01, FR-02, FR-12, NFR-03, NFR-11
Logging / Audit	Infrastructure	Records significant AI and user events with details, timestamps, and confidence where applicable.	NFR-08
Model Monitoring	Infrastructure	Uses inference errors, confidence values, feedback, and latency observations to support later quality review.	FR-03, FR-07, FR-12, NFR-05, NFR-08
Environment	External	Phone camera, operating system, gym equipment, research API connectivity, and video services.	FR-03, FR-04, FR-06, FR-07, NFR-01, NFR-02

6. Design Questions and Decisions

DQ1. How should the system handle incorrect equipment recognition?

The result view displays the predicted equipment and confidence. Low-confidence output remains provisional and the user is asked to change the angle, distance, or lighting. The confidence sheet exposes alternative classes and allows confirmation or rejection. Manual search provides a non-AI fallback, while feedback and prediction details are written to the audit log.

DQ2. Should equipment recognition run on-device or on a remote server?

Recognition runs on the phone through TensorFlow Lite. This keeps camera data local, removes network latency from the core scan path, and preserves equipment identification when connectivity is unavailable. The packaged model and label file are loaded once by the detection service.

DQ3. How fast should equipment recognition be to keep the system usable?

The interaction target is a result within three seconds. The camera screen uses a live preview and triggers image capture and inference periodically, preventing concurrent inference requests. The trained classifier recorded an average inference latency of 9.63 ms on the CUDA evaluation system; end-to-end mobile response also includes capture, decoding, resizing, and device-specific interpreter time.

DQ4. How should the system behave when the camera cannot detect equipment?

After 15 seconds without a reliable result, the interface requests a camera or lighting adjustment and presents manual search. Camera errors are handled without closing the application, and the user can retry, change cameras, or enable flash where supported.

DQ5. How can the system reduce hallucinations in research summaries?

Retrieval precedes generation. The system selects up to three local research records and inserts their titles, citations, and abstracts into the prompt. The prompt restricts the answer to the supplied material and prohibits invented statistics. Source cards remain visible under the generated summary.

DQ6. How should the system prevent unsafe fitness advice?

The prompt asks for concise, practical, research-bounded information. Generated content is checked against unsafe terms before display. Flagged responses are replaced with a safety message, and every

result includes a statement that the content is not medical advice and that professional guidance may be required.

DQ7. How should user privacy be protected when collecting workout history and goals?

Sessions, sets, goals, preferences, and audit records are stored in the local SQLite database. Recognition is performed on-device and raw camera images are not retained. The profile contains an AI-recommendation toggle and a function that deletes workout data, goals, and audit history.

DQ8. How should personalised recommendations be generated without excessive complexity?

The prototype uses a transparent content-based method. Each fitness goal maps to relevant muscle groups; the equipment and exercise services filter the local catalogues accordingly. This method works with a single user and does not require population-level interaction data.

DQ9. How should the system support users when no internet connection is available?

Scanning, manual catalogue search, workout logging, goal selection, and dashboard calculations use local assets and storage. Research generation and external video playback require connectivity. If generation is unavailable, the research screen presents the retrieved source excerpts instead of an unsupported answer.

DQ10. How should the system monitor AI quality over time?

The audit table records CV labels, mapped equipment, confidence, user confirmations or denials, research-summary events, and fallback events. These records provide evidence for class-specific review, model updates, threshold adjustment, and diagnosis of recurring failures.

DQ11. How should manually searched and camera-detected equipment share the same instruction flow?

Both paths resolve to the same Equipment entity from the bundled catalogue. The selected record is then passed to the same equipment card and instruction screen. This removes duplicated content and keeps descriptions, aliases, tips, videos, and workout logging consistent.

DQ12. How should the system explain AI outputs to users?

Equipment results include confidence and alternative predictions. Research responses are marked as AI and RAG output and list the retrieved sources. Recommendation content is displayed alongside the selected fitness goal, making the basis of the suggestion visible rather than presenting it as an unexplained ranking.

7. Implementation Overview

The delivered prototype is organised as a Flutter application with screen, service, model, widget, and utility packages. Application state is intentionally lightweight: singleton services load bundled assets and coordinate SQLite operations, while individual screens own transient UI state. The source tree separates AI inference from catalogue lookup, persistence, and presentation logic.

Area	Implementation
Client framework	Flutter 3.x / Dart
Computer vision runtime	TensorFlow Lite through tflite_flutter
CV architecture	ResNet-50, 23-class equipment classifier
Research generation	Gemini 2.0 Flash through the Google Generative AI SDK
Retrieval	Local JSON knowledge base with keyword-based ranking
Persistence	SQLite (sqflite) and SharedPreferences
Charts	fl_chart
Camera	Flutter camera package
External content	Instructional video URLs opened with url_launcher

7.1 Computer Vision Pipeline

The classifier is bundled as `assets/equipment_classifier.tflite` with a label file containing 23 classes. Camera images are decoded, resized to 224 x 224 pixels, converted to RGB floating-point values in the range [0, 1], and passed to the interpreter in NHWC layout. Output scores are sorted to produce the leading prediction and alternatives. The configured acceptance threshold is 0.75. A successful label is resolved through the equipment catalogue so that both camera and manual-search flows use the same domain record.

7.2 Research Retrieval and Summarisation

The research knowledge base is stored in `assets/research.json`. Each record contains a title, authors, publication year, abstract, and retrieval keywords. The service scores records against the user query and supplies the top three to the generative model. Generation uses a low temperature and a short output limit to favour concise, source-bounded responses. The screen displays the generated text, the source citations, and an explicit AI/RAG label. When generation is unavailable, the selected research excerpts remain accessible.

7.3 Local Data Model

SQLite table	Stored information
workouts	Equipment sessions, duration, notes, and timestamp.
workout_sets	Exercise, set number, weight, repetitions, notes, and timestamp.
goals	Current goal type and optional target period.
audit_log	Event type, details, confidence, and timestamp.
settings	GDPR consent and AI-recommendation preference.

Dashboard metrics are calculated from these tables rather than stored as duplicated values. Personal records use the Epley estimated one-repetition maximum, allowing sets with different repetition counts to be compared on a common basis.

7.4 Error Handling and Fallbacks

- The scan feature reports camera initialisation errors, low confidence, no-detection timeout, and inference failure without terminating the app.
- Manual search remains available when recognition is unreliable or the equipment class is unsupported.
- The research screen handles missing sources, network failure, invalid API configuration, and quota errors with user-readable messages.
- If the bundled model cannot be loaded, the scan screen can demonstrate the interaction through a controlled demo path.
- Local write operations are isolated in the database service, and destructive deletion requires confirmation in the profile screen.

8. Computer Vision Model Evaluation

The final equipment classifier was evaluated on a held-out test split after training and validation. The training set contained 10,326 cropped equipment images, the validation set 976 images, and the test set 504 images. Model selection used validation macro-F1, which limits bias toward classes with more examples.

Metric	Result
Architecture	resnet50
Number of classes	23
Training / validation / test	10326 / 976 / 504
Best validation macro-F1	0.9533
Test loss	0.1586
Test top-1 accuracy	0.9603 (96.03%)
Test top-3 accuracy	0.9881 (98.81%)
Test macro precision	0.9521
Test macro recall	0.9543
Test macro-F1	0.9513
Average inference latency	9.63 ms on the CUDA evaluation device

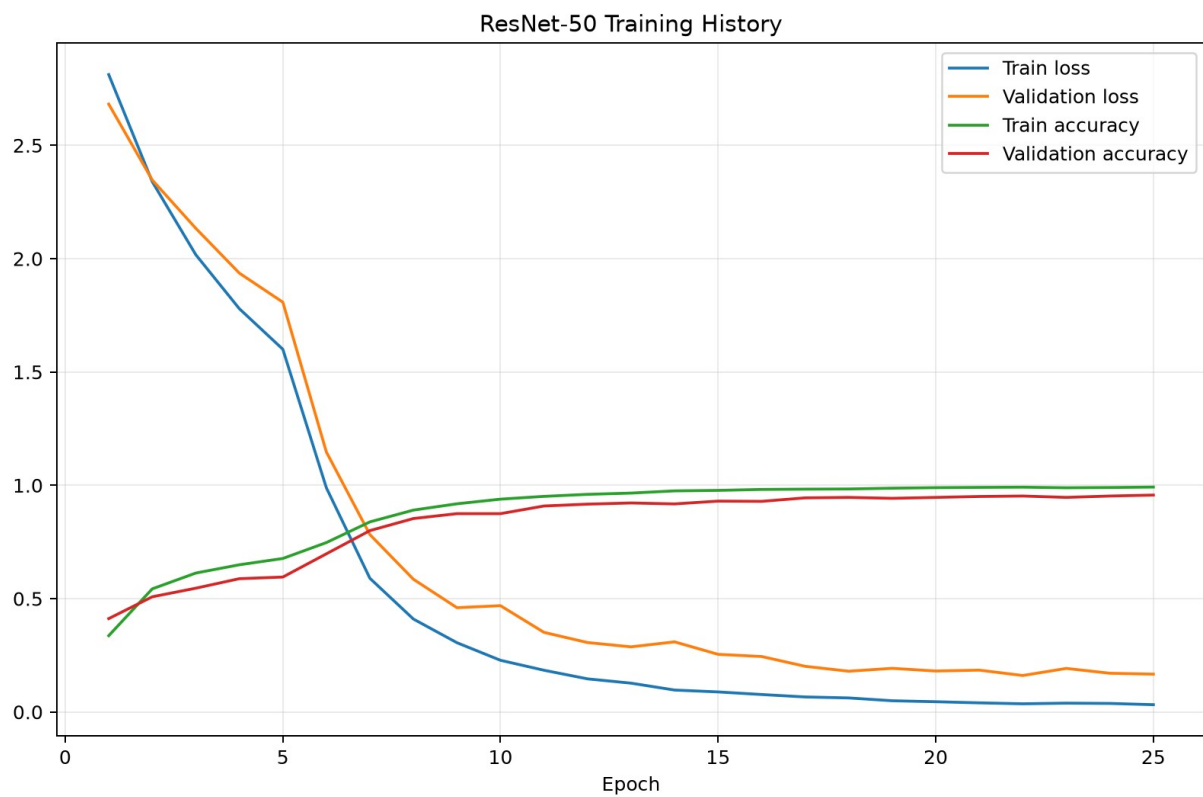


Figure 4. ResNet-50 training and validation loss and accuracy.

Training and validation loss decreased consistently, while validation accuracy stabilised near the final test performance. The small gap between the curves after fine-tuning indicates controlled overfitting rather than a large train-test divergence.

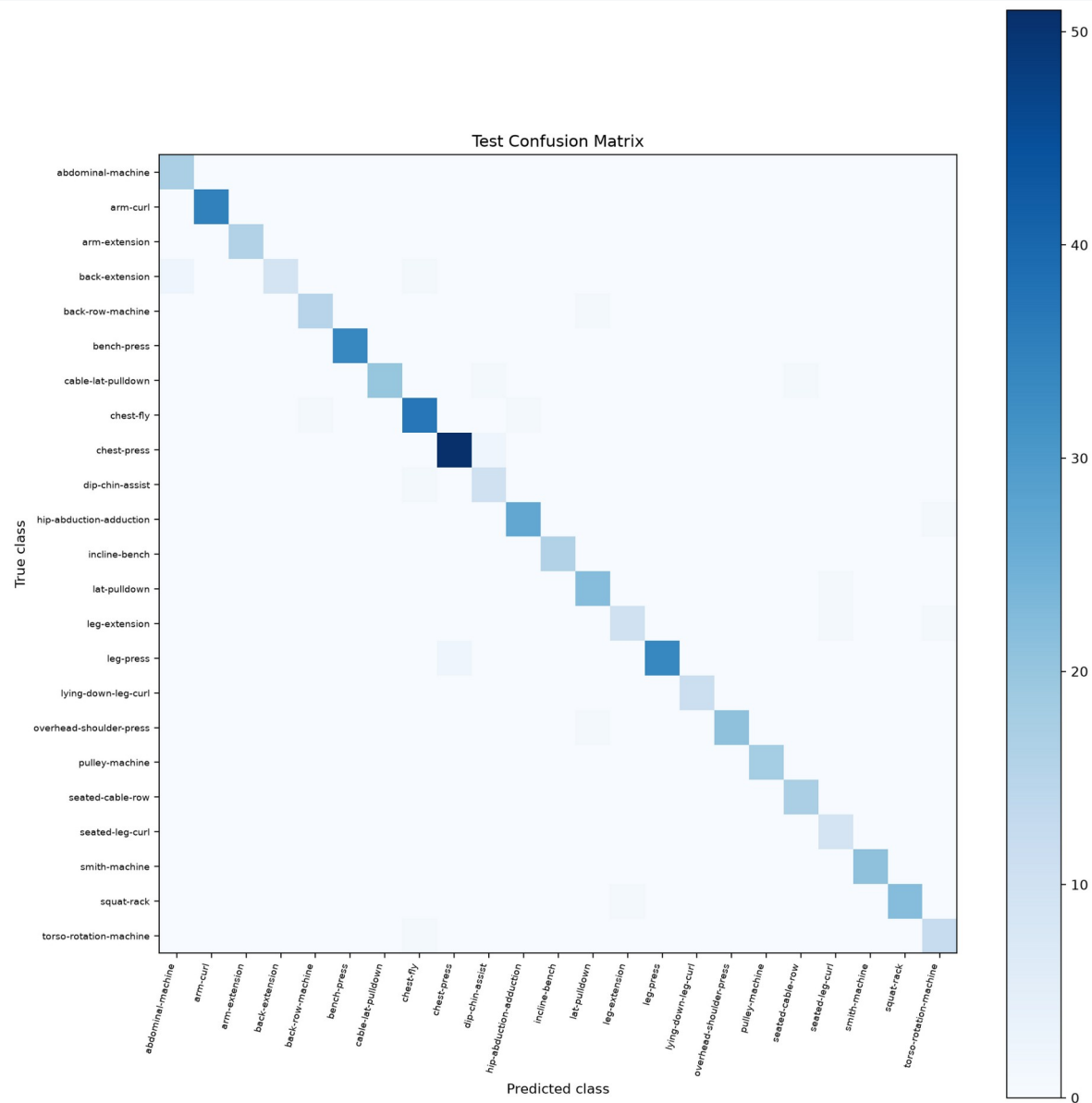


Figure 5. Test-set confusion matrix for the 23 equipment classes.

The confusion matrix is strongly diagonal. Several classes achieved perfect test precision and recall, including arm curl, arm extension, bench press, incline bench, lying-down leg curl, pulley machine, and smith machine. The most difficult classes were back extension, dip/chin assist, leg extension, seated leg curl, and torso rotation, where visually similar machine frames and body positions produced occasional cross-class predictions.

8.1 Supported Equipment Classes

Class	Class	Class
abdominal-machine	arm-curl	arm-extension
back-extension	back-row-machine	bench-press
cable-lat-pulldown	chest-fly	chest-press
dip-chin-assist	hip-abduction-adduction	incline-bench
lat-pulldown	leg-extension	leg-press
lying-down-leg-curl	overhead-shoulder-press	pulley-machine
seated-cable-row	seated-leg-curl	smith-machine
squat-rack	torso-rotation-machine	

8.2 Evaluation Limits

The test results describe performance on cropped images from the same dataset distribution used for model development. Mobile use introduces additional variation from distance, background clutter, partial visibility, manufacturer design, camera orientation, and lighting. The application therefore treats the confidence threshold, top alternatives, user confirmation, and manual search as part of the recognition design rather than relying on the top-1 class alone.

9. Conclusion

GymGeek implements the eight principal use cases defined for the project: authentication, camera-based equipment identification, instructional content, research summarisation, workout logging, dashboard presentation, manual search, and fitness-goal management. The architecture keeps camera inference and personal workout data local, while network-dependent functions are isolated behind service interfaces and accompanied by fallbacks.

The prototype demonstrates how an AI-intensive mobile system can combine predictive components with ordinary application logic, traceable requirements, explicit safety controls, and user-facing explanations. The computer vision model exceeded the specified top-3 target on the held-out test data, and the application integrates its output with a shared equipment catalogue rather than treating model predictions as standalone results.

End of document